QUALITY PROBLEM SOLVING · TEXTILES

Applying Japan's PQPS Methodology: Cutting a Textile Finishing Defect by 60%

Structured QC Circle Mentoring — Fakhruddin Textile Mills Ltd, Urmi Group

4.7% → 1.87%	~$1,510	Platinum
DEFECT RATE	REPROCESSING SAVINGS	NATIONAL QC AWARD

THE SITUATION

At Fakhruddin Textile Mills Ltd (Urmi Group), the finishing division was losing product to "peach uneven" — an inconsistent surface texture on mechanically peach-finished fabric caused by uneven sanding, roll to roll or meter to meter. Over a 5-month baseline, this defect averaged 4.7% of output, driving reprocessing costs and delivery risk on production for brands including M&S, Puma, and Decathlon.

WHAT I DID

Rather than leaving the shop floor to troubleshoot ad hoc, I introduced the structured quality problem-solving methodology I trained in through AOTS Japan's PQPS (Program for Quality Problem Solving), and mentored a 4-person QC Circle team through the full cycle:

1. **Structured project selection** — guided the team to brainstorm five candidate defects, then score each on cost, feasibility, and importance using a weighted matrix, objectively prioritizing "peach uneven" as the top issue over four alternatives.
2. **Root cause mapping** — directed root-cause analysis using a fishbone (Man / Method / Material / Machine) diagram and a T-diagram breakdown, surfacing nine potential causes.
3. **Pareto-driven focus** — had the team score and rank all nine causes, then applied Pareto analysis to show that two causes — peach roller fiber length variation and fabric tension variation — accounted for 76% of the problem, focusing effort where it mattered most.
4. **Countermeasure design (5W1H)** — guided the team to define specific, assigned countermeasures: operator care protocols for roller fiber, a 2-month roller replacement cycle, fixed fabric tension at the first roll, and switching to flat-lock sewing for roll-to-roll joins.
5. **PDCA governance** — structured the full initiative on a Plan-Do-Check-Act cycle across roughly 3.5 months, with defined phases for root-cause analysis, solution testing, and standardization.
6. **Standardization** — once verified, converted the countermeasures into a formal SOP and ran floor-level training for all peach machine operators and helpers to sustain the gains.

THE RESULT

- ✓ Defect rate dropped from a 4.7% average to 1.87% within 3 months of implementation
- ✓ Reprocessing cost savings of approximately 128,400 Taka (~$1,510) over the measured period
- ✓ The QC Circle's work was independently recognized with a Platinum Award at the 24th National Annual Quality Convention (NAQC), Bangladesh Society for Total Quality Management

WHY IT MATTERS

This reflects the ability to bring formal, internationally-trained quality methodology into a real production environment — not just applying tools, but building the structure and mentoring a team to use them independently and sustainably.